

Sprint 3

Simulation Engine

Full Sprint Review | April 17, 2026

Deloitte Capstone — AI for Equitable Public Transportation

University of Miami

Agenda

- 1 Sprints 1 & 2 Recap + Sprint 3 Overview
- 2 Sprint 3.1: Baseline State & Lever Catalog
- 3 Sprint 3.2: Simulation Engine
- 4 Transition to Client-Ready Dashboard
- 5 Live Dashboard Demo
- 6 Conclusion & Limitations

Items 4 & 5 are conducted as a live dashboard demo outside this slide deck

Project Journey: Sprints 1 > 2 > 3

Sprint 1 — EDA

DONE

- GTFS feed analysis: 128 routes, 6,530 stops, 943K stop-times
- ACS 2023 Census data: 707 tracts, 56 demographic variables
- Initial equity landscape of Miami-Dade transit

Sprint 2 — Modeling

DONE

- 7 equity indicators, multiplicative composite score (Need x Deficit)
- XGBoost v3: $R^2 = 0.823$, 13 features, leakage-free
- ACS time-series: 2027 need projections, fragile tract flagging

Sprint 3 — Simulation

CURRENT

- 504-tract baseline assembled with equity tiers + risk flags
- 4 transit-service levers, simulator.py engine, 16/16 tests
- Client-ready interactive dashboard for policy scenarios

Baseline State Assembly

504 census tracts joined from 3 source files into a single baseline for the simulator:

- **Risk Scores (v3)**
 - XGBoost deficit predictions + residuals + risk tiers
- **Need Projections (2027)**
 - ACS time-series forecasts + fragile/worsening flags
- **Equity Indicators (v3)**
 - 7 indicators, composite scores, tier assignments

504

Census Tracts

13

Model
Features

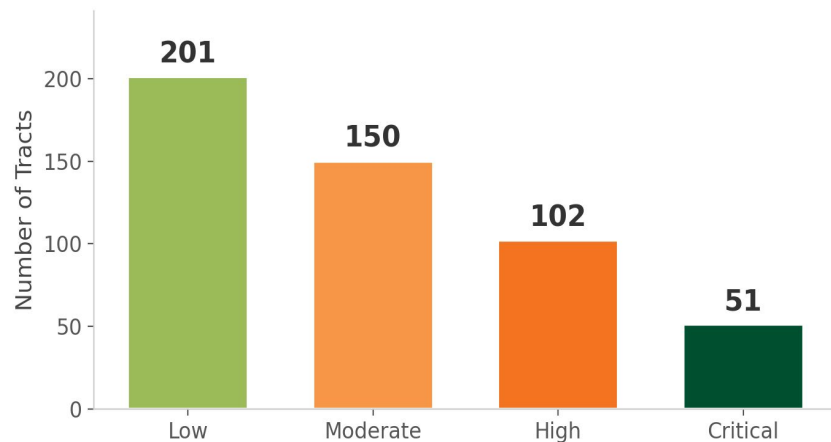
22

Fragile Tracts

120

Worsening
Tracts

Equity Tier Distribution — 504 Census Tracts



The 4 Transit-Service Levers

Lever	What It Means	SHAP Rank	Baseline Avg	Range
Peak AM Frequency	Buses/trains per hour during morning rush	#1 (39.6%)	26.7 tph	0 – 228
Weekend Ratio	Weekend service as fraction of weekday service	#2 (21.7%)	0.43	0 – 1.3
Early AM Frequency	Trips/hr before 6 AM (shift workers, early commuters)	#3 (6.9%)	3.0 tph	0 – 28
Rail Trip Share	Fraction of trips on rail (modal shift from bus)	Largest what-if impact	1.1%	0 – 79%

These are parameters a transit agency can change by signing a work order — not demographic forecasts.

Rail Trip Share has the largest what-if impact: +10pp → -0.0149 deficit, 98% of tracts improve | Peak AM Frequency is #1 by SHAP importance (39.6%)

How the Simulator Works

1

Adjust Levers

User changes transit service parameters (e.g. +2 trips/hr peak AM)

2

Re-Predict Deficit

XGBoost v3 model estimates new service deficit per tract

3

Update Equity = Need x Access Deficit

Proportional change ratio applied to equity priority scores

4

Report Results

Tier shifts, tract-level impacts, summary statistics generated

Need side (demographics) is held static — simulator only adjusts what transit planners can control

Dependencies & Architecture

- XGBoost v3 model (Sprint2b_XGBoost_v3.pkl) — CV $R^2 = 0.823$
- Sprint3_Baseline_State.csv — 504 tracts, all features + targets
- Sprint3_Lever_Catalog.json — 4 levers with valid ranges + SHAP ranks
- Headway auto-derived from frequency inside simulator (60/freq)
- Zero-frequency tracts: headway capped at 120 min (training convention)

Validation & Scenario Results



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test_simulator.py::TestZeroDelta::test_equity_unchanged PASSED [ 6%]
test_simulator.py::TestZeroDelta::test_deficit_unchanged PASSED [ 12%]
test_simulator.py::TestZeroDelta::test_no_tier_shifts PASSED [ 18%]
test_simulator.py::TestZeroDelta::test_summary_zeros PASSED [ 25%]
test_simulator.py::TestIsolation::test_unaffected_unchanged PASSED [ 31%]
test_simulator.py::TestIsolation::test_affected_changed PASSED [ 37%]
test_simulator.py::TestDirection::test_freq_peak_am PASSED [ 43%]
test_simulator.py::TestDirection::test_weekend_ratio PASSED [ 50%]
test_simulator.py::TestDirection::test_rail_share PASSED [ 56%]
test_simulator.py::TestDirection::test_freq_early PASSED [ 62%]
test_simulator.py::TestStructural::test_invalid_lever PASSED [ 68%]
test_simulator.py::TestStructural::test_tract_count PASSED [ 75%]
test_simulator.py::TestStructural::test_result_shape PASSED [ 81%]
test_simulator.py::TestStructural::test_tier_shift_matrix PASSED [ 87%]
test_simulator.py::TestStructural::test_equity_bounds PASSED [ 93%]
test_simulator.py::TestStructural::test_shap_ranking PASSED [100%]
```

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===== 16 passed in 0.79s =====
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Validation & Scenario Results

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16/16 Pytest Gates Pass



Zero-Drift (4 tests)

No change in levers = no change in output. Baseline perfectly recovered.



Isolation (2 tests)

Targeting specific tracts only affects those tracts. No spillover.



Direction (4 tests)

Every lever moves deficit in the expected direction per SHAP rankings.



Structural (6 tests)

API contracts, edge cases, empty filters, tier assignment logic.

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PASSED [ 12%]
PASSED [ 18%]
PASSED [ 25%]
PASSED [ 31%]
PASSED [ 37%]
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PASSED [ 75%]
PASSED [ 81%]
PASSED [ 87%]
PASSED [ 93%]
PASSED [100%]
```

From Model to Action

Client-Ready Dashboard

We will now switch to the live dashboard to demonstrate:

1. How to use the Policy Simulator with real scenarios
2. Actionable insights for Miami-Dade transit planning

Demo Scenario

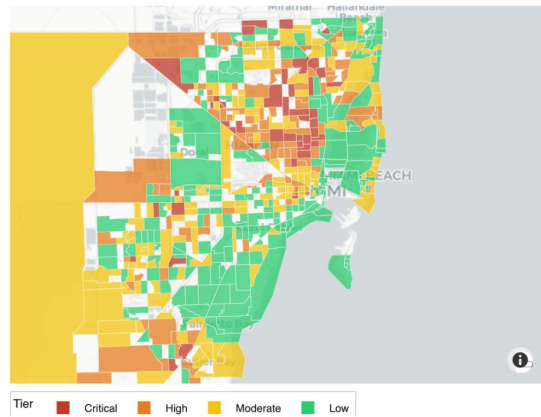
S1: Peak Frequency Boost

+2 trips/hr during AM peak, targeting Critical tracts (51)

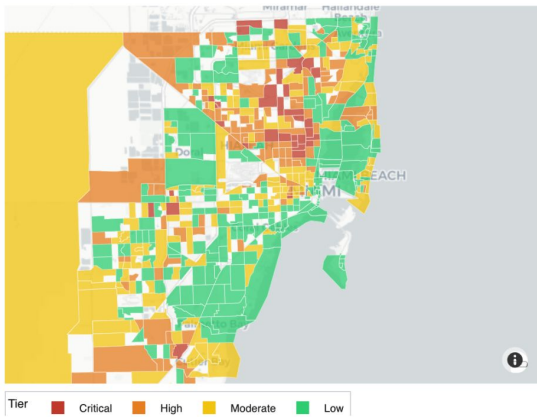
Expected: ~51 critical tracts improve, 15 tier upgrades, 0 downgrades

Insight: Shows how morning rush improvements ripple through equity scores

Before



After



FROM TO	LOW	MODERATE	HIGH	CRITICAL
Low	200	-	-	-
Moderate	-	150	-	-
High	-	-	102	-
Critical	-	-	15	37

Lina + Daniel Dashboard Walkthrough

What We Delivered

Sprint 2a

Measure the Gap

7 equity indicators across 504 Miami-Dade census tracts. Composite score: Need x Access Deficit. 4-tier classification.



Sprint 2b

Model the Fix

XGBoost v3 predicts service deficit ($R^2 = 0.823$). 13 leakage-free features. SHAP-ranked policy levers. ACS 2027 need forecasts.



Sprint 3

Simulate the Solution

General-purpose simulator engine. 4 transit-service levers. 16/16 validation tests. Interactive client-ready dashboard.

From raw data to actionable transit policy recommendations

Limitations & Next Steps

Current Limitations

- **Not a constrained optimization model**

We were not given constraints (budget, resources, human capital). The simulator shows impact of policy choices, not the optimal allocation.

- **Need side held static in v1**

Demographic projections (poverty, vehicle access) are fixed forecasts. Stress-test scenarios deferred to v2.

- **No spatial spillover**

Improving one tract does not yet propagate benefits to neighboring tracts.

With Constraints, We Would...

- Retrain models with budget/resource constraints as features
- Feed dashboard outputs back to recalculate optimal allocations
- Build constrained optimization layer (maximize equity improvement per dollar)
- Add need-side demographic stress-test scenarios (v2)
- Integrate graph features from city2graph as model inputs

This simulator is a scenario analysis tool that shows the direction and magnitude of impact. With real-world constraints, it becomes an optimization engine.